

Project Workflow

Project Workflow: Week 8

Phase: AI Opponent & UX Polish

1. **Objective:** The primary goal of this week was to complete the planned work for AI Opponent & UX Polish. This included:
 - Implement a basic AI paddle behavior for single-player mode.
 - Improve the user interface with a clean, modern appearance and visual feedback.
 - Optimize the Timer interval for smooth and fluid gameplay.
2. **Detailed Task Breakdown:**
 - **AI Opponent:** Implemented basic AI tracking logic based on ball y -position for Player 2.
 - **UI Polish:** Added anti-aliasing rendering, modern colors, and distinct scores for Player vs AI.
 - **Timer Optimization:** Reduced timer interval to 14ms (~70 FPS) for smooth movement.
 - **Gameplay Review:** Checked AI reaction speeds and visuals during active gameplay.
3. **Deliverables:**
 - Basic AI opponent behavior.
 - Polished game user interface.
 - Visual feedback improvements.
 - Optimized Timer interval for smooth gameplay.
4. **Tools & Resources:**
 - Programming Language: Java
 - GUI Framework: Java Swing
 - Game Loop/Timing: Timer
 - Development Environment: IntelliJ IDEA / VS Code

Project Workflow: Week 9

Phase: Error Handling & Testing

1. **Objective:** The primary goal of this week was to complete the planned work for Error Handling & Testing. This included:
 - Add exception handling for invalid states and edge cases.
 - Test paddle and ball collision under different speeds.
 - Fix bugs and optimize the overall game performance.
2. **Detailed Task Breakdown:**
 - **Exception Handling:** Embedded `try-catch` blocks inside the game loop timer to safely recover from runtime exceptions.

- **Collision Testing:** Adjusted directional bounces and positional clamps to prevent the ball from passing through paddles.
 - **Bug Fixing:** Fixed ball sticking bugs along top and bottom window boundaries.
 - **Performance Optimization:** Enhanced thread safety by running Swing frame setup using `SwingUtilities.invokeLater`.
3. **Deliverables:**
- Exception handling for invalid states and edge cases.
 - Tested paddle and ball collision behavior.
 - Fixed gameplay bugs.
 - Improved overall game performance.
4. **Tools & Resources:**
- Programming Language: Java
 - GUI Framework: Java Swing
 - Testing: Gameplay and collision testing
 - Development Environment: IntelliJ IDEA / VS Code

Project Workflow: Week 10

Phase: Final Testing, Documentation & Submission

1. **Objective:** The primary goal of this week was to complete the planned work for Final Testing, Documentation & Submission. This included:
 - Perform complete gameplay testing.
 - Prepare project documentation and screenshots.
 - Finalize and submit the Ping Pong Game project.
2. **Detailed Task Breakdown:**
 - **Complete Testing:** Tested game states including Win (Player reaching 10 points), Game Over (AI reaching 10 points), Pause (SPACE), and Restart (R).
 - **Documentation:** Prepared full source code documentation and workflow breakdown.
 - **Final Review:** Verified all deliverables from Week 1 through Week 10 are completely met.
 - **Submission:** Finalized code clean-up for final project submission.
3. **Deliverables:**
 - Fully tested Ping Pong Game with Win/Game Over conditions.
 - Project documentation.
 - Finalized project ready for submission.
4. **Tools & Resources:**
 - Programming Language: Java
 - GUI Framework: Java Swing
 - Development Environment: IntelliJ IDEA / VS Code
 - Testing: Complete gameplay testing